District-Level Rice Yield Prediction in Odisha Using Time-Aware Machine Learning Regression

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***Abstract*—Accurate crop yield forecasting is central to food security, resource planning, and agricultural policy, particularly in states such as Odisha where rice is the dominant kharif crop and district-level planning depends on reliable production estimates. This paper presents a machine-learning framework for predicting district-level rice yield (kg/ha) in Odisha using eleven years (2014–2024) of integrated agricultural, climate, soil, nutrient, and irrigation data. Two district/season/crop-wise government datasets were merged on fiscal year, district code, crop, and season, restricting the analysis to rice and to district-level (non-aggregate) records, which produced 869 matched observations and, after removing missing targets, 839 records suitable for supervised learning. Unlike many crop-yield studies that rely on a random train/test split, this work treats yield forecasting as a genuine future-prediction problem: models are trained on 2014–2021 (665 records) and evaluated on a chronological, previously unseen 2022–2024 holdout (174 records), with model selection performed using five-fold, year-aware forward cross-validation rather than random k-fold validation. Four regression algorithms — Linear Regression, Decision Tree Regressor, Random Forest Regressor, and Gradient Boosting Regressor — were compared inside a leakage-safe preprocessing pipeline (standardized numerical features, one-hot encoded categorical features), with Random Forest and Gradient Boosting additionally tuned via time-aware randomized search. The tuned Random Forest Regressor was selected as the final model on the basis of the lowest mean cross-validation RMSE (639.40 kg/ha) and achieved a test RMSE of 541.77 kg/ha, a test MAE of 435.91 kg/ha, and a test R^2 of 0.4165 on the 2022–2024 holdout. Feature-importance analysis, residual analysis, and error breakdowns by year, season, and district were used to interpret the model, and an important data-quality issue — an unseen season label ("Kharif") appearing only in the test period — was identified and documented as a limitation. In contrast to studies that report near-perfect accuracy using random splits on small, low-noise datasets, this paper's moderate explanatory power ($R^2 = 0.4165$) reflects a more realistic and honestly reported estimate of what is achievable when yield prediction is evaluated under a genuine future-year, out-of-time protocol. To improve transparency, the final pipeline is further examined using SHAP global explanations, LIME instance-level explanations, and robustness checks. A Streamlit dashboard is also implemented and deployed to expose the saved model through overview, district analysis, prediction, explainability, and model-performance views. The dashboard is an interface to the research model and does not change its predictive performance. The study is positioned as a research/academic model rather than a production decision system, and concrete directions for improving accuracy — larger and longer time series, finer spatial resolution, and additional agronomic variables — are outlined.**

***Index Terms*—Agricultural machine learning, rice yield prediction, Random Forest, time-aware cross-validation, chronological train/test split, precision agriculture, Odisha, SHAP, LIME, explainable AI, Streamlit dashboard, regression.**

I. Introduction

Rice is the principal kharif crop of Odisha and a cornerstone of the state's agrarian economy, supporting the livelihood of a large share of its rural population. Year-to-year and district-to-district variation in rice yield, driven by rainfall, temperature, soil nutrient status, and irrigation coverage, creates uncertainty for farmers, procurement agencies, and policymakers alike. Machine learning offers a data-driven alternative to traditional trend-based or expert-judgment yield estimates by learning non-linear relationships between environmental, soil, and agronomic variables and observed yield outcomes.

A recurring weakness in published crop-yield machine-learning studies, however, is evaluation methodology: many report accuracy from a random train/test split, which allows information from future years to leak into model training and can produce misleadingly optimistic performance figures. Because yield forecasting is, in practice, a future-year prediction problem — a model trained on past seasons is used to predict a season that has not yet occurred — evaluation should mirror that constraint. This paper adopts a chronological train/test split and a time-aware, year-forward cross-validation scheme so that no validation year is ever used to inform predictions for an earlier year.

This study integrates two publicly available government statistical sources describing district-, season-, and crop-wise area, production, and yield for Odisha together with environmental and agricultural covariates (rainfall, temperature, humidity, soil pH, nitrogen, phosphorus, potassium, and irrigation coverage), and builds a complete, auditable machine-learning pipeline: data integration and validation, exploratory data analysis (EDA), leakage-safe preprocessing, chronological splitting, time-aware model selection and hyperparameter tuning, final evaluation, and residual/error analysis.

The contributions of this study are summarized as follows:

1) **Dataset integration and validation**: two independent district-level Odisha agricultural datasets are merged on fiscal year, district code, crop, and season, with state-aggregate records removed and the target variable recomputed directly from production and area to correct inconsistent unit reporting in the source data.
2) **A realistic, leakage-aware evaluation protocol**: a chronological 2014–2021 / 2022–2024 train/test split combined with five-fold year-aware forward cross-validation, in contrast to the random-split evaluation common in comparable literature.

3) **A systematic comparison of four regression algorithms** (Linear Regression, Decision Tree, Random Forest, Gradient Boosting) with randomized hyperparameter search restricted to the training period, and model selection based on validation RMSE rather than test performance.
4) **Model interpretation and transparent limitation reporting**, including feature importance, residual and error analysis by year/season/district, and documentation of a season-label distribution shift ("Kharif") between the training and test periods.
5) **Explainability and robustness analysis**: global SHAP analysis, four representative LIME instance explanations, and sensitivity/consistency checks to assess the stability and agreement of explanations.
6) **Accessible model delivery**: implementation and public deployment of a Streamlit dashboard that uses the saved preprocessing-and-model pipeline for district analysis, yield prediction, model explainability, and performance inspection.

The remainder of this paper is organized as follows. Section II reviews related work on machine-learning approaches to crop classification and yield prediction. Section III describes the dataset, feature engineering, preprocessing pipeline, and modelling methodology. Section IV presents and discusses the results, feature importance, and error analysis. Section V concludes the paper and outlines directions for future work.

II. Related Work

A substantial body of work has applied machine learning to crop recommendation and yield prediction. Badshah et al. [1] proposed two machine-learning pipelines for agricultural decision-making: a crop recommendation classifier trained on a 2,200-record Kaggle dataset of soil and weather attributes (achieving 99.7% accuracy with Random Forest across seven classifiers), and a wheat-yield regression model for Pakistan built on World Bank/FAO data from 1992–2013 and extended to 2014–2025 using Multivariate Imputation by Chained Equations (MICE), for which a tuned Support Vector Regressor reported an R^2 of 99.9%. Explainable-AI techniques — LIME and feature-importance analysis — were used in both pipelines to interpret model behaviour. While methodologically thorough in its feature engineering and explainability treatment, the wheat-yield component relies on synthetically imputed future-year data and does not report a strictly out-of-time (future-year) holdout evaluation, which limits direct comparability with genuine forecasting performance.

De Clercq and Mahdi [2] investigated district-level Kharif rice-yield prediction across 247 Indian rice-producing districts using 20 years of climate reanalysis and satellite data, training 19 models (including CatBoost, LightGBM, and Extremely Randomized Trees) and reporting out-of-sample R^2 up to 0.82, together with SHAP-based feature attribution identifying temperature and leaf area index as dominant drivers. Paidipati et al. [3] applied Support Vector Regression with linear, polynomial, and RBF kernels to long-run (1962–2018) national and state-level Indian rice yield series, evaluating kernels via cross-validated RMSE and MAE. Sahoo et al. [4] compared Cubist, Gradient Boosting Machine, MARS, Random Forest, SVM, and XGBoost for Kharif rice yield-gap prediction in eastern India under CMIP6 climate scenarios, reporting R^2 of 0.72–0.73 for the best-performing Cubist and Random Forest models and near-negligible R^2 for several other learners on the same features — underscoring that algorithm choice and feature set interact strongly with achievable accuracy. Sah et al. [5] combined optical and SAR remote-sensing data with crop biophysical parameters to predict rice yield at 45, 60, and 90 days after transplanting, finding that prediction accuracy improved as the observation date approached harvest and that XGBoost, neural-network, and Cubist models performed best at different growth stages.

Relative to this literature, the present study differs in two respects. First, most comparable rice-yield studies report performance from random cross-validation or random train/test partitions; this study instead enforces a strict chronological split and a year-aware forward validation scheme, which generally produces more conservative — but more decision-relevant — accuracy estimates. Second, the study documents, rather than conceals, a real data-quality issue (an unseen categorical season label appearing only in the test period) and its effect on model behaviour, in line with recent calls in the literature for greater transparency in agricultural machine-learning evaluation.

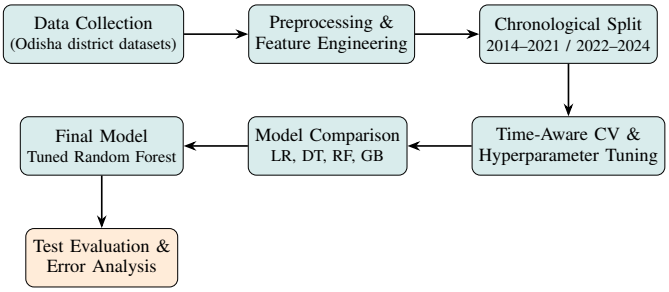


Fig. 1. Block diagram outlining the proposed Odisha rice-yield prediction methodology.

III. Methodology

The overall workflow, from raw district-level data through to test-set evaluation, is summarized in Fig. 1.

A. Data Collection and Integration

Two publicly available district-level datasets for Odisha are used. The first is a district-, season-, and crop-wise area, production, and yield statistics dataset; the second is an integrated dataset of environmental and agricultural variables covering rainfall, temperature, humidity, soil pH, nitrogen, phosphorus, potassium, and irrigation coverage. The two sources are merged using four shared keys — fiscal year, district LGD (Local Government Directory) code, crop, and season. Only Rice observations are retained, and 'State Total' aggregate rows are removed prior to merging because the study focuses exclusively on district-level observations. The inner merge yields 869 matched district-level observations; after removing

records with missing target values, 839 observations remain for supervised learning.

B. Target Variable

The prediction target is rice yield in kilograms per hectare, `yield_kg_ha`, computed directly as:

$$\text{Yield (kg/ha)} = \frac{\text{Production (tonnes)} \times 1000}{\text{Area (ha)}} \quad (1)$$

This derived value is used in place of the dataset's raw `reported_crop_yield` field because the source data contains inconsistent unit representation across records, and recomputing the target directly from production and area removes this inconsistency.

C. Exploratory Data Analysis

Exploratory analysis covers dataset structure and data types, missing-value and duplicate checks, district-name validation, merge validation, target-distribution and outlier analysis (using the IQR rule), year-wise observation counts, yield trends over time, seasonal and district-level yield comparisons, climate and soil feature distributions, feature-versus-yield relationships (including area, rainfall, and temperature trends), soil pH and N/P/K analysis, irrigation analysis, correlation analysis, and an area-versus-production validation check. Outliers are deliberately not removed automatically, since unusually high or low agricultural yields may reflect genuine district-level conditions rather than data errors. The `production` field is retained for validation and EDA but excluded from the modelling feature set because it is mathematically related to the target and would otherwise cause target leakage.

D. Features

The final feature set comprises ten numerical variables: `area` (hectares under rice cultivation), `year`, `avg_temperature_c`, `rainfall_mm`, `humidity_pct`, `soil_ph`, `nitrogen_kg_ha`, `phosphorus_kg_ha`, `potassium_kg_ha`, and `irrigation_pct`; and two categorical variables: `district` and `season`.

Six fields are explicitly excluded from the feature set: `yield_kg_ha` (the target itself), `production` (target leakage, since yield is derived from it), `reported_crop_yield` (a redundant, target-derived field), `district_lgd_code` (a non-informative identifier), `district_source` (a redundant district-name field), and `fiscal_year` (redundant with `year`). Fig. 2 summarizes how the retained input features relate to the predicted output.

E. Preprocessing Pipeline

All preprocessing is implemented inside a scikit-learn `ColumnTransformer` wrapped in a single `Pipeline` together with the estimator, so that scaling and encoding statistics are fit only on training folds during cross-validation, preventing data leakage. Numerical features are standardized (zero mean, unit variance) using `StandardScaler`; categorical features (`district`, `season`) are transformed with `OneHotEncoder(handle_unknown="ignore")`, so that any category unseen during training does not cause the pipeline to fail at inference time. This preprocessing expands the 12 raw predictors into 43 features after one-hot encoding.

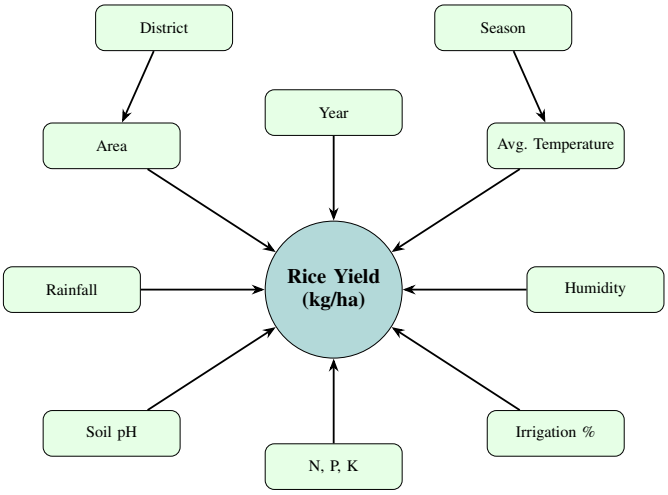


Fig. 2. Block diagram showing input features and the predicted output variable.

TABLE I
CHRONOLOGICAL TRAIN/TEST SPLIT OF THE ODISHA RICE-YIELD DATASET

Dataset	Years	Observations
Training	2014–2021	665
Final Test	2022–2024	174
Total	2014–2024	839

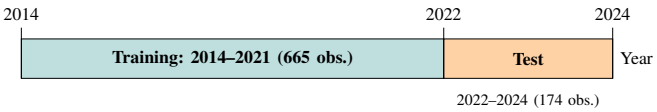


Fig. 3. Chronological train/test split used for time-aware evaluation.

F. Chronological Train/Test Split

Rather than a random split — which would allow the model to be validated on years chronologically earlier than some of its training data — the 839 observations are partitioned strictly by year, as shown in Table I.

This split reflects the intended real-world use case: models are trained on historical years and used to predict future, previously unseen years, as illustrated in Fig. 3.

G. Time-Aware Cross-Validation and Hyperparameter Tuning

Model selection within the training period (2014–2021) uses year-aware forward validation: validation folds keep complete years together, and a given validation year is always chronologically after the years used to train that fold, mirroring the eventual forecasting task. Five time-aware folds are used. Hyperparameters for Random Forest and Gradient Boosting are tuned using `RandomizedSearchCV` applied only to the

training period; the 2022–2024 test set is held out entirely from model selection and hyperparameter tuning and is used only once, for final evaluation.

H. Candidate Models

Four regression algorithms are compared: Linear Regression (as a baseline linear model), Decision Tree Regressor, Random Forest Regressor, and Gradient Boosting Regressor. Random Forest and Gradient Boosting are ensemble tree methods that combine many weak learners — via bootstrap aggregation and sequential residual-fitting, respectively — to reduce variance and capture non-linear interactions among climate, soil, and agronomic features. Model selection is based on the lowest mean validation RMSE across the five time-aware folds, with validation MAE used as a secondary criterion; final test-set performance plays no role in choosing the winning model.

I. Explainable-AI Analysis

The selected Random Forest pipeline is examined using SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations). SHAP is used to study feature contributions globally and at the instance level. The Tree-Path-Dependent SHAP formulation is used as the primary explanation method for the tree ensemble. Explanations are generated for both the training and chronological test partitions, and their rankings are compared to assess whether the dominant model signals remain similar across the two periods. Encoded-feature explanations are also grouped back to their original input variables to make the results easier to interpret. SHAP explanations are treated as descriptions of model behaviour, not as causal effects of agricultural variables.

LIME is applied to four representative test observations to provide local approximations around individual predictions. The cases include an underprediction for Bargarh (Winter 2022), an overprediction for Deogarh (Summer 2024), and two Kharif-period observations from Kendrapara (2024) and Boudh (2023). The local explanation scores differ substantially across cases; therefore, individual LIME explanations are interpreted with caution, especially where local fidelity is low. For the Kharif observations, the unseen season category is retained as recorded rather than being silently relabelled.

J. Explanation Robustness Checks

Additional checks assess the stability of the global SHAP rankings and the sensitivity of LIME explanations to random seed, number of samples, and kernel width. Agreement between SHAP and LIME is assessed only for eligible feature-direction comparisons. The resulting agreement rate is reported with its comparison count, rather than being interpreted as universal agreement between the methods. These checks are intended to characterize explanation stability; they do not constitute an independent validation of predictive accuracy.

K. Streamlit Dashboard Implementation

A Streamlit application is developed as an interactive interface to the completed model. The dashboard is organized into five views: Overview, District Analysis, Yield Prediction, Model Explainability, and Model Performance. It loads the serialized preprocessing-and-model pipeline from the project artifacts and uses that saved pipeline for inference; it does not retrain the estimator when a user opens the application. The application is deployed through Streamlit Community Cloud and is linked from the project GitHub repository. This deployment makes the research workflow easier to inspect and demonstrate, but it should not be interpreted as evidence of improved model accuracy or as a production-grade agricultural advisory service.

TABLE II
FINAL PERFORMANCE OF THE TUNED RANDOM FOREST REGRESSOR ON THE 2022–2024 HOLDOUT

Metric	Result
Selection CV RMSE	639.40 kg/ha
Test RMSE	541.77 kg/ha
Test MAE	435.91 kg/ha
Test R^2	0.4165

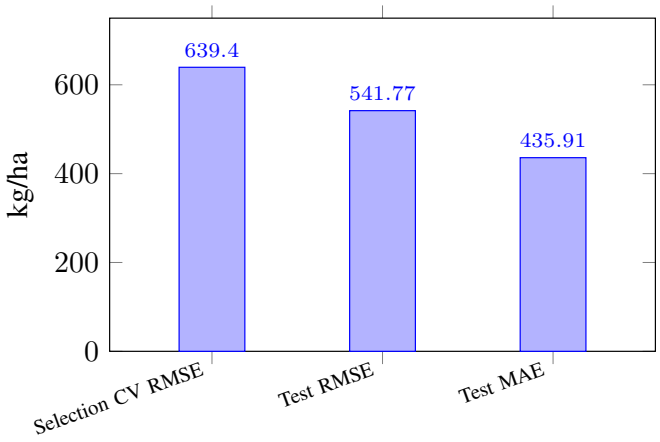


Fig. 4. Final error metrics of the tuned Random Forest Regressor (Test $R^2 = 0.4165$).

IV. RESULTS AND DISCUSSION

The tuned Random Forest Regressor was selected as the final model on the basis of the lowest mean time-aware cross-validation RMSE. Its performance on the training-period validation folds and on the held-out 2022–2024 test period is summarized in Table II.

The final model is deliberately selected using training-period cross-validation rather than by choosing whichever model performs best on the final test set, avoiding an implicit form of test-set leakage that is common when model choice is influenced by holdout performance. Fig. 4 visualizes these three error metrics side by side.

A. Model Interpretation

Feature-importance analysis, prediction-versus-actual plots, residual analysis, absolute-error analysis, and error breakdowns by year, season, and district are used to interpret the final model, alongside a comparison of training and test

performance to check for overfitting. Consistent with standard practice, feature importance here is interpreted strictly as a measure of model reliance (how much the Random Forest uses a given feature to reduce prediction error), not as evidence of a causal relationship between that feature and rice yield.

The SHAP analysis uses the Tree-Path-Dependent formulation as the primary explanation method. The expected value is 2177.20 kg/ha, and the SHAP arrays have dimensions of 665 by 43 for training and 174 by 43 for testing, corresponding to the transformed feature space. The maximum additivity gaps are approximately 7.28×10^{-12} for training and 4.55×10^{-12} for testing, indicating numerical agreement between the SHAP decomposition and model outputs. The five leading grouped input features are season, year, district, area, and humidity. The top-five overlap between the training and test rankings is 5/5, with a Spearman rank correlation of 0.9161. The grouped-input and encoded-feature analyses also have a top-five overlap of 5/5 and a rank correlation of approximately 0.993. An interventional SHAP formulation was not selected as the primary result because its maximum additivity gap was 39.1675 (mean gap 3.3245), substantially larger than that of the Tree-Path-Dependent formulation.

LIME explanations are generated for four selected test observations using quartile discretization, 5,000 perturbation samples, 12 displayed features, and a fixed random seed of 42. The local R^2 values vary from 0.0296 for the Bargarh Winter 2022 underprediction to 0.6718 for the Deogarh Summer 2024 overprediction; the two selected Kharif cases have local R^2 values of 0.1424 and 0.1447. These values indicate that local surrogate fidelity is uneven and can be weak for some observations. In the robustness checks, SHAP/LIME feature-direction agreement is 82.6% across 46 eligible comparisons. The consistency checks passed 10 out of 10 predefined verification checks. However, top-five feature overlap between SHAP and LIME for the four individual cases ranges from 2/5 to 4/5, reinforcing that global rankings are more stable than every instance-level explanation. The robustness results support cautious interpretation rather than a claim that all explanations are interchangeable.

B. Deployed Dashboard

The deployed Streamlit dashboard provides five user-facing sections: (i) Overview for a summary of the dataset and project, (ii) District Analysis for exploring district-level patterns, (iii) Yield Prediction for obtaining predictions from user-supplied inputs, (iv) Model Explainability for inspecting the available interpretation outputs, and (v) Model Performance for reviewing evaluation metrics. The dashboard loads the saved model artifact and preprocessing pipeline, ensuring that inference uses the same fitted transformations as the evaluated model. The source code and project documentation are maintained in the public GitHub repository (https://github.com/kalpana-sketch/Rice_Yield_Prediction), and the application is hosted on Streamlit Community Cloud (https://riceyieldprediction-wgowxbc5trijckm4fai95q.streamlit.app/). The dashboard is intended for demonstration and research communication; it does not remove the limitations of the underlying dataset or model.

C. Comparison with Related Work

The test R^2 of 0.4165 is markedly lower than the near-perfect accuracies reported for the wheat-yield and crop-recommendation models in Badshah et al. [1] (R^2 up to 99.9%, accuracy up to 99.7%), but is broadly consistent with — and in fact somewhat below — the 0.72–0.82 range reported by district- and regional-scale rice-yield studies that use random or spatial cross-validation [2], [4]. This gap is attributable less to modelling choices than to evaluation methodology and data volume: this study enforces a strict chronological, out-of-time holdout on a comparatively small sample (839 observations across 11 years), which is a substantially harder and more realistic test than a random split on a larger or lower-noise dataset. The result illustrates a broader pattern in the agricultural machine-learning literature: reported accuracy is highly sensitive to whether evaluation is random or time-aware, and R^2 values above 0.99 on agricultural yield data, as reported elsewhere, typically reflect either abundant, low-noise engineered data or a random split rather than a genuine future-year forecasting test.

D. Data-Quality Finding: Season-Label Drift

An important limitation surfaced during error analysis. The training data (2014–2021) contains only three season labels — Autumn, Summer, and Winter — while the test period (2022–2024) additionally contains a fourth label, Kharif, affecting 60 test observations. Because the categorical encoder is configured with `handle_unknown="ignore"`, the pipeline continues to run without error, but observations bearing the unseen Kharif label receive no learned season-specific signal, since no corresponding one-hot column exists from training. This is flagged as a genuine limitation of the current dataset rather than a modelling defect, and is a concrete direction for future feature engineering — for example, mapping Kharif to its agronomically closest equivalent among the training labels, or encoding season using a continuous representation (e.g., sowing/harvest month) instead of a categorical label.

E. Limitations

Several limitations affect the interpretation and generalizability of the results:

1) **Limited dataset size**: 839 observations is relatively small for learning detailed district- and season-specific patterns.
2) **Limited temporal history**: only 2014–2024 is available, with just three years (2022–2024) used as the final holdout.
3) **Season-label drift**: the Kharif label appears in the test period but not in training, as discussed above.
4) **District-level aggregation**: environmental and soil variables are available at district level rather than at finer spatial resolutions such as blocks or individual farms.
5) **Missing explanatory variables**: the dataset does not include crop variety, pest/disease incidence, exact sowing or harvest dates, extreme-weather events, or more detailed soil characteristics.

6) **Future-year uncertainty**: the test set covers only three future years relative to the training period, so performance beyond 2024 is not established.
7) **Moderate explanatory power**: a test R^2 of 0.4165 means a substantial share of yield variation in the holdout remains unexplained by the model.

For these reasons, the model presented here should be treated as a research and academic prediction model rather than a production agricultural decision-support system.

V. Conclusion and Future Scope

This paper presented a complete, leakage-audited machine-learning workflow for predicting district-level rice yield in Odisha, integrating government agricultural statistics with climate, soil, nutrient, and irrigation data across 2014–2024. In contrast to much of the comparable literature, evaluation was performed using a strict chronological train/test split and year-aware forward cross-validation, so that reported performance reflects a genuine future-year forecasting scenario rather than a random split. Among four candidate regressors, a tuned Random Forest Regressor was selected using training-period cross-validation and achieved a test RMSE of 541.77 kg/ha, a test MAE of 435.91 kg/ha, and a test R^2 of 0.4165 on the 2022–2024 holdout. Feature-importance, SHAP, and LIME analyses were used to interpret model behaviour, with additional robustness checks documenting both stable global rankings and less consistent local explanations. A season-label distribution shift between the training and test periods was identified and documented as a concrete limitation rather than concealed. A Streamlit dashboard was also implemented and deployed to provide access to district analysis, prediction, explainability, and model-performance views using the saved model pipeline.

The comparatively modest R^2, when read alongside the near-perfect accuracies often reported for similarly framed agricultural prediction tasks, underscores a methodological point relevant to the wider field: evaluation protocol — chronological versus random splitting, in particular — has a larger effect on reported accuracy than algorithm choice, and honestly reported, time-aware performance figures are more useful to practitioners than optimistic figures obtained under random cross-validation.

Future work can extend this study along several directions:

1) Extending the temporal range of the dataset beyond 2024 and beyond 2014 where historical records permit, to give the model more years of variation to learn from and a longer future-year test window.
2) Resolving the season-label drift (Kharif vs. Autumn/Summer/Winter) through agronomically informed relabeling or a continuous, date-based encoding of the growing season.
3) Incorporating finer-grained spatial data (block- or farm-level) and additional agronomic variables such as crop variety, sowing/harvest dates, and pest or disease incidence.
4) Conducting further external and temporal validation on additional years and, where possible, on independently collected district-level datasets to assess transferability beyond the current holdout.
5) Improving the deployed dashboard with additional validated data sources, clearer uncertainty communication, and user feedback, while retaining its role as a research demonstration rather than an operational advisory system.

Overall, the study demonstrates that a transparent, leakage-aware, time-respecting machine-learning pipeline is achievable for district-level rice-yield prediction in Odisha, and that documenting a model's real limitations is as valuable a contribution as the predictive accuracy figure itself.

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